

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		At least 2 satellites indicated which could be used for communicating between A and B, e.g. at 2 o'clock and 4 o'clock (1) Additional ground station indicated which can communicate with the satellites (1) Lines up and down A → satellite → ground station → satellite → B (1) If more than 2 satellites drawn, the system must be capable of working to award any and each mark.		3		3		
		(ii)		[A base station can be in constant communication with a geostationary satellite] because it always stays in the same position above the Earth (1) but only once every 24 h with a geosynchronous satellite (1)	2			2		
	(b)			Doubling of distance = $2 \times 3.6 \times 10^4$ [= 7.2×10^4 km] (1) Substitution: 3×10^8 [m/s] = $\frac{7.2 \times 10^4 [\text{km}]}{\text{time}}$ (1) ecf on not doubling distance. Manipulation and conversion: $\text{time} = \frac{7.2 \times 10^7}{3 \times 10^8}$ (1) Time = 0.24 [s] [1] Answer of 0.12 s (3) NB 24×10^n ($n \neq -2$) → 3 marks 12×10^n ($n \neq -2$) → 2 marks	1	1 1 1		4	4	
				Question 5 total	3	6	0	9	4	0